

Adv C Module

1. Subjective

2.1 Basic Refreshers

1. What is C language, and what are its key features?
2. What are the basic data types in C?
3. What is the difference between int, float, and double?
4. What are variables and how are they declared?

2.2 1D Pointers and Functions

1. What is a pointer?
2. What are the different types of pointers?
3. What is the difference between pass by value and pass by reference?

2.3 String

1. What is a string in C, and how is it stored in memory?
2. What is the difference between a character array and a string literal?
3. What are commonly used string functions (strlen, strcpy, strcmp, etc.)?
4. What is the difference between:

```
char str[] = "hello";  
char *str = "hello";
```

2.4 Storage classes and memory segments

1. What are the storage classes in C? Explain each (auto, register, static, extern).
2. What is the difference between static and external variables?
3. What are the different memory segments in a C program?
4. Where are the following stored:
 - a. Global variables
 - b. Local variables
 - c. Static variables
 - d. Dynamically allocated memory
5. What is the difference between stack and heap memory?

2.5 2D Pointers and DMA

1. What is the difference between malloc() and new?
2. What is the difference between a global variable and a DMA pointer?
3. What is void, Dangling and Wild pointer?
4. What is memory leakage? How to avoid it?
5. What is a segmentation fault?
6. What are function pointers?

2.6 Preprocessing

1. What are the stages of compilation?
2. What is the role of the C preprocessor?
3. What are macros, and how are they defined using #define?
4. What is the difference between:
 #include <file.h>
 #include "file.h"
5. What are conditional compilation directives (#if, #ifdef, #ifndef)?
6. What are function-like macros, and what are their advantages and pitfalls?

2.7 UDT

1. What is a structure, and how is it different from an array?
2. What is the difference between a structure and a union?
3. What is the purpose of typedef in C?
4. What are nested structures? Give an example.
5. How do you pass a structure to a function (by value vs by pointer)?

2.8 Miscellaneous

1. What is a segmentation fault, and when does it occur?
2. What is memory leakage, and how can it be avoided?
3. What is undefined behavior? Give examples.

2.9 FILE I/O

1. What is a file pointer (FILE *) in C?
2. What are the different file opening modes (r, w, a, etc.)?
3. What are binary file operations (fread, fwrite)?

2. Programming

1. WAP to Given two strings, determine whether they are isomorphic.
2. Write a program to check whether a given number is a power of 2.
3. Given two arrays of sizes N and N-1, find the missing element.
4. WAP to swap two numbers using bitwise operators?
5. Given input 0xABCD, how do you transform it to 0xBADC using bitwise operations?
6. WAP to implement atoi().
7. WAP to implement itoa().
8. Write a C program to swap two numbers without using a temporary variable.
9. Write a program to find the largest number in an array.
10. Write a C program to print alternate characters from two strings.
11. Write a program to read a file line by line and print only the lines that contain a given word.
12. WAP to replace a specific word in a file with another word.
13. Write a program to find a file in a directory and print its path.
14. Write a program to extract specific bits from an integer using bitwise operations.
15. Write a program to check whether it is a perfect power (c^d).
16. Write a program to count special numbers from 1 to N, where a number is not a perfect power.
17. WAP to rotate number: 12345678 $\rightarrow$ 56781234.
18. How do you swap two numbers using:
 - a. Pass by value
 - b. Pass by reference
19. An encryption system processes an integer array of size N, but due to a system failure, some elements are replaced with -1, indicating missing values. You are given the array A. Replace each -1 with a valid 4-bit binary number such that the number is even (its least significant bit is 0) and greater than or equal to 4. Among all such values, choose the smallest possible number. After replacing all -1 values, compute and return the sum of the updated array.
20. A building has N floors and M elevators. Each elevator operates between a specific start floor and a stop floor. You are given two arrays: Arr1[] representing the starting floors of the elevators and Arr2[] representing the corresponding stopping floors. Each elevator

can only move between its defined start and stop floors. Starting from floor 1, determine the highest floor that can be reached using any combination of these elevators.

21. You are given a number N in the form of a string. A number is considered a special number if it cannot be expressed in the form $c^d \cdot d^c$, where $c > 1$ and $d > 1$ (i.e., it is not a perfect power). Count how many numbers from 1 to N (inclusive) are special numbers, and return the result as a string.

MC Module

1. Basic electronics

1. What is a resistor?
2. What is a capacitor?
3. What is an inductor?
4. What is a diode?
5. What is a transistor?
6. What is the difference between a capacitor and an inductor?
7. What is the purpose of a pull-up resistor?
8. What is the purpose of a decoupling capacitor?
9. How do you measure resistance?
10. How do you test a capacitor?
11. How do you test a diode?
12. How do you measure voltage?
13. How do you measure current?
14. What is a multimeter, and how is it used?
15. What is the difference between a series circuit and a parallel circuit?
16. What is ground in a circuit?
17. What is a voltage regulator?

2. Basics

1. How did you interface the RTC module?
2. How did you store data in EEPROM?
3. What type of power supply did you use?
4. What hardware challenges did you face, and how did you solve them?
5. What is an RC circuit?
6. What is the difference between a MOSFET and a BJT?
7. What is the difference between an RL circuit and an RC circuit?

3. Interrupts

1. What is an interrupt handler?
2. What are timers and how are they used?
3. What is a watchdog timer and why is it important?
4. What are interrupts? Explain their working.
5. If the processor is executing an interrupt handler and a higher-priority interrupt occurs, what will happen?

4. Projects

1. Explain the car black box project.
2. Can you draw and explain the block diagram?
3. Explain your CAN-based automotive dashboard project.

5. ADC

1. What is an ADC, and why is it used in embedded systems?
2. What is the difference between resolution and accuracy in ADC?
3. What is sampling rate, and why is it important?
4. What are different types of ADCs (SAR, Flash, Sigma-Delta)?
5. How do you interface an ADC with a microcontroller? Explain the steps.

6. Embedded Systems

1. What is an embedded system? Give examples.
2. What is the difference between a microcontroller and a microprocessor?
3. What are the key components of an embedded system?
4. What is real-time processing, and what is the difference between hard and soft real-time systems?

7. Protocols

1. What is the I2C protocol? Explain how it works.
2. What is the SPI protocol? Explain its working.
3. What are inter-board and intra-board communication protocols?
Explain with examples.
4. Explain the I2C protocol.
5. What is the difference between 7-bit and 10-bit addressing in I2C?
How do they work?
6. How does the SPI protocol work?
7. How many wires are used in I2C?
8. What is a start condition?
9. What is a stop condition?
10. What is ACK/NACK?
11. What is the difference between master and slave?
12. Why are pull-up resistors used in I2C?
13. What does full duplex communication mean?
14. Explain master-slave communication in SPI.
15. What is the purpose of the chip select (CS) line?
16. What is the difference between SPI and I2C?
17. What is baud rate?
18. What are start and stop bits?
19. What is the difference between UART and SPI?

8. PWM

1. What is PWM, and why is it used?
2. What is duty cycle, and how does it affect output?
3. What is the relationship between frequency and PWM signal?
4. How is PWM generated in a microcontroller?
5. What are practical applications of PWM (e.g., motor control, LED dimming)?

CPP Module

1. What is STL?
2. What is inheritance?
3. What is an abstract class?
4. What is a pure virtual function?
5. What is the difference between a structure in C and a class in C++?
6. What is an object?
7. What is function overriding?
8. What is the difference between C pointers and C++ pointers?
9. What is runtime vs compile-time polymorphism?

Ds Module

1. Basics

1. What is a data structure, and why is it important?
2. What is the difference between linear and non-linear data structures?
3. What is time complexity and space complexity?
4. What is Big-O notation? Give examples.

2. Linked lists

1. WAP to delete a node in a doubly linked list?
2. WAP to reverse a singly linked list?
3. WAP to find the loop in a single linked list.
4. WAP to reverse linked list using recursion.
5. WAP to swap the last two nodes.
6. Write a program to remove a loop in a linked list.

3. Stack

1. What is a stack, and what are its operations?
2. What is the difference between stack using array vs linked list?
3. What is overflow and underflow in stack?
4. Write an algorithm to check balanced parentheses using stack.
5. What are applications of stack (e.g., recursion, expression evaluation)?

4. Searching and Sorting Techniques

1. WAP to implement Merge Sort.
2. What is the difference between linear search and binary search?
3. What are different sorting algorithms?
4. What is the time complexity of:
 - a. Bubble sort
 - b. Selection sort
 - c. Merge sort
5. How does binary search work? What are its constraints?
6. What is the difference between stable and unstable sorting algorithms?

5. Queue

1. What is a queue, and what are its operations?
2. What is the difference between:
 - a. Linear queue
 - b. Circular queue
3. What is overflow and underflow in a queue?
4. What is a priority queue?
5. What are applications of queue (e.g., scheduling, buffering)?

6. Hashing

1. What is hashing, and why is it used?
2. What is a hash function? What makes it good?
3. What are collisions, and how are they handled?
4. What is the difference between:
 - a. Chaining
 - b. Open addressing
5. What is the load factor in hashing?

7. Trees

1. What is a tree data structure?
2. What are different tree traversals (inorder, preorder, postorder)?
3. What is the height of a tree, and why is it important?
4. What is a balanced tree, and why do we need it?

LI Module

1. Basics

1. What are different CPU scheduling algorithms?
2. Write a command or program to replace a word in a file without opening it manually.
3. Write a command to find a file in the current directory.
4. Write a command to check running processes in the system.
5. Write a command to kill a process by PID.

2. System call

1. What is a system call, and why is it required?
2. What are the different types of system calls (process, file, device, information)?
3. What is the difference between a library function and a system call?
4. Explain fork(), exec(), and wait() system calls.
5. What happens internally when a system call is invoked?

3. Networking

1. What is the OSI model, and what are its layers?
2. What is the difference between TCP and UDP?
3. What is an IP address, and what is the difference between IPv4 and IPv6?
4. What is a port number, and why is it used?
5. What is a 3-way handshake in TCP?

4. Process

1. What is a process, and how is it created?
2. What are the different states of a process?
3. What is a Process Control Block (PCB)?
4. What is context switching?
5. What is the difference between a process and a program?

5. IPC

1. What is IPC, and why is it needed?
2. What are different IPC mechanisms (pipe, message queue, shared)?
3. What is the difference between pipe and FIFO?
4. What is shared memory, and how does it work?
5. What is a semaphore, and how is it used for synchronization?

6. Signal

1. What is a signal in Unix/Linux?
2. What are common signals (SIGINT, SIGKILL, SIGTERM)?
3. What is a signal handler?
4. What is the difference between signal() and sigaction()?
5. Can signals be ignored or blocked? Explain.

7. Project

1. Explain the TFTP project.
2. Explain the minishell project.

8. Socket

1. What is a socket, and how is it used in communication?
2. What are the steps to create a server socket?
3. What are the steps to create a client socket?
4. What is bind(), listen(), accept() in socket programming?

9. Threads

1. What is a deadlock? How can it be prevented?
2. What is a thread, and how is it different from a process?
3. What are user-level and kernel-level threads?
4. What is thread synchronization?
5. What are mutex and semaphore in threads?
6. What is race condition, and how can it be avoided?

10. Process and Memory Management

1. What is virtual memory, and why is it used?
2. What is paging, and how does it work?
3. What is a page fault?
4. What is memory allocation (static vs dynamic)?
5. What is fragmentation (internal and external)?

Qt Module

1. What is Qt, and what is its usage?
2. What is QMediaPlayer?
3. What is the signal and slot mechanism in Qt? How does it work?
4. What is the role of the Qt Event Loop, and how does event handling work in Qt applications?

General Questions Aptitude

1. Where do you see yourself in the next five years?
2. Why did you apply for this role?
3. What differentiates you from other candidates who have similar skills?
4. What is your hometown known for?
5. Can you introduce yourself from a hardware perspective?
6. Are you more interested in hardware or software? Why?
7. Can you explain the hardware architecture of your project?
8. Can you describe the block diagram of your project?
9. What debugging tools have you used?
10. State Ohm's Law.
11. What is Kirchhoff's Current Law (KCL)?
12. What is Kirchhoff's Voltage Law (KVL)?
13. What is the difference between series and parallel resistors?
14. What is the voltage divider formula?
15. How do you handle project deadlines?
16. How do you handle pressure?
17. Are you willing to relocate?